

The PCMark CPU index shows a variation of about 300 points across all the DDR boards. This shows us how the chipset implementation can directly impact performance numbers even while using the same processor. The lowest score of 5012 went to the Gigabyte GA-8S MML, which is a SiS 650-based DDR solution.

The best performing SDRAM board in this category was the Intel 845 chipset-based Mercury KOB845NFSX, which managed to score a respectable PCMark CPU index of 5233.

As you will see in our Scoreboard table, minor but significant optimisations that manufacturers like MSI and ASUS incorporate clearly show up in the SiSoft Sandra CPU Multimedia benchmarks. All the Intel chipset based boards score about 8600 odd points in the CPU Integer benchmark. The MSI 845 Ultra-C and the MSI 845 Ultra-ARU however score 8779 and 8758 respectively, which proves that they squeeze every last bit of performance out of the Pentium 4. Even the original Intel D850MD RDRAM-based board is relegated to scoring only 8673. A similar trend was observed in the FPU SSE benchmarks where the MSI board outshone all others.

Athlon XP/Duron: The competition in this category was fierce with the top boards scoring 4900 and above in the PCMark CPU index. The ASUS A7V266-E scored a hefty 4957 and the DFI AD70-SC scored an impressive 4908, but they couldn't touch the nForce-based ASUS A7N266-E. It was the only AMD board which breached the 5000 barrier—it notched an amazing score of 5006.

The pure integer crunching strength of the AMD boards could be seen in the

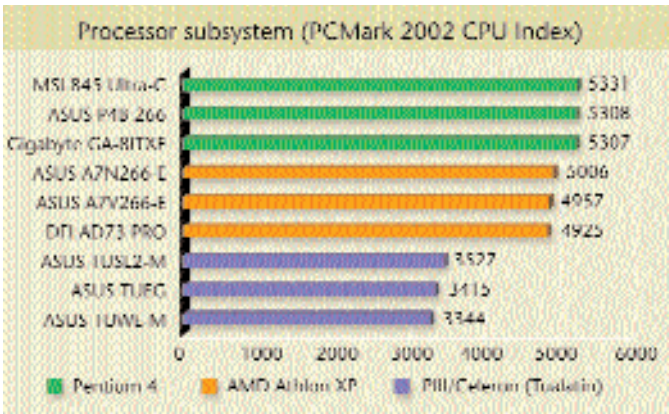
Which Chipset is for me?			
	Pentium 4*	Pentium III/Celeron**	Athlon XP/Duron
SDRAM	Intel 845	Intel 810E2, Intel 815EG, Intel 815EP, SiS 630ET	VIA KT133A, VIA KM133A, VIA KLE133
DDR SDRAM	Intel 845WN, Intel D845BG, Intel 845D, SiS 650, SiS 645, VIA P4M266, P4X266A	Not available	VIA KT333, VIA KT266A, nVidia nForce 420-D, SiS 735
RDRAM	Intel 850GB, Intel 850MD	Not available	Not available

**This refers to Socket 478 processors only **This refers to the Tualatin processors only
The table reflects only the chipsets that we've tested*

SiSoft Sandra CPU Integer benchmark as almost every board scored over 9000 compared to the relatively lower scores of the Pentium 4 boards which hovered around

liantly. The ASUS TUSL2-M was particularly spectacular, scoring 3527 in the PCMark CPU index. This was more than 100 points over the older ASUS TUEG

which scored just 3415. The ASUS TUSL2-M is a Micro ATX board which performs brilliantly with both the Intel Pentium III and the Celeron processors.



8600. The Soltek SL75DRV4 took top honours, beating the ASUS A7V266-E by about 6 points, with a score of 9210. A score of above 9000 should ensure adequate performance, while anything in excess of 9150 is really impressive.

Pentium III/Celeron (Tualatin): This category is dominated by ASUS with their latest Tualatin boards performing bril-

liantly. To squeeze every last drop of performance, the memory subsystem must have very tight memory timings and implementation.

Despite the presence of strict memory standards, there will be slight variation in the memory modules produced by different manufacturers. The motherboard has to ensure that the memory slots can han-

Memory sub-system
The memory sub-system is the most critical because it needs to constantly feed the proces-

A Handy Checklist

Buying a new motherboard? Here's our checklist for what you need to look out for: Maximum CPU speed supported: This can save you a lot of heartache if you plan on upgrading your CPU at a later date. You can get this information from the motherboard's manual or from the manufacturer's Web site.

Number of PCI slots: More PCI slots give you greater flexibility in case you decide to add PCI devices later. As a rule, look for a minimum of three PCI slots, though six slots would be ideal.

More memory slots: While most motherboards come with two memory slots, having a third or fourth RAM slot may come in handy if you need to add more RAM later. USB 2.0: You should look for at least four USB ports as USB is soon becoming the interface of choice for most PC peripherals. Support for ATA-133: With hard disks getting faster, ATA-133 provides that much needed bandwidth for applications such as video editing and computer games. Hardware monitoring: This is one feature that you simply cannot do without as

processor speeds (and running temperatures) keep rising. In case the CPU fan fails, these motherboards will automatically shut down the system and prevent damage to the core components. Warranty and support: Check whether the dealer is providing at least a two-year replacement warranty and has a dedicated RMA centre in India. Regular BIOS updates: Regular BIOS updates eliminate bugs that might have cropped up after the motherboard was shipped.